



Existential threat and compliance with pro-environmental norms

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ABSTRACT

Recent research on the effects of personal threat suggests that perceived threat might enhance pro-environmental behavior when pro-environmental norms are in focus. In three experiments we found support for the latter assumption, showing that mortality salience and salience of pro-environmental norms interacted in predicting pro-environmental attitudes and information search (Study 1), sustainable behavior in a forest management game (Study 2), as well as pro-environmental intentions and behavior (Study 3). Specifically, mortality salience increased pro-environmental conduct only when pro-environmental norms were salient. Moreover, norm salience only had an effect on pro-environmental attitudes and behavior when the threat of personal mortality was salient. We discuss the implications of these results for both terror management theory and the promotion of pro-environmental behaviors.

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1. Introduction

In most industrialized countries prescriptive norms of pro-environmental conduct have become increasingly important and culturally shared. However, non-ecological behaviors are still widespread in the everyday lives of most people, such as driving small distances with large vehicles or putting toxic substances in the dustbin. Particularly, under conditions of threat and societal uncertainty, it is often argued that ecological standards and norms are becoming less important than in times of perceived wealth and societal stability. As in many industrialized countries, increasing economic globalization has led to a rise in perceptions of personal insecurity and unpredictability (e.g., regarding personal risks such as long-term unemployment or being insufficiently protected by the health care system), one may assume that compliance to pro-environmental norms might decrease. However, on the other hand, perceptions of individual (e.g., Fritsche & Kessler, *Unpublished manuscript*) or collective threat (e.g., Duckitt & Fisher, 2003) have been shown to result in defense of ingroup norms and worldviews. Therefore, in countries where pro-environmental norms are deeply anchored in society and are thus salient in everyday life, perceptions of self threat might lead to even more of an inclination to act in a pro-environmental manner. In this article we deal with these contradictory predictions by testing the hypothesis that the

experience of personal threat can increase adherence to pro-environmental norms and thus pro-environmental behavior, but only when these norms are salient in a situation. By doing so we combine both norm focus theory (Cialdini, Kallgren, & Reno, 1991) and terror management theory (e.g., Greenberg, Solomon, & Pyszczynski, 1997) and investigate the impact of perhaps the most basic personal threat one can think of: The threat of personal death.

1.1. Pro-environmental norms and their impact on behavior

Pro-environmental norms are social norms which are generally defined as “rules and standards that are understood by members of a group, and that guide and/or constrain social behavior without the force of laws” (Cialdini & Trost, 1998, p. 152). However, although the influence of social and personal norms on pro-environmental behavior is highlighted in the pertinent literature (e.g., Bamberg & Möser, 2007; Stern, 2000), the relationship between social norms of environmental protection and individuals’ behaviors is far from perfect. Therefore, the investigation of conditions under which the perception of pro-environmental social norms indeed results in norm consistent behavior, is an important research agenda in environmental psychology. Building on predictions from the norm activation model (Schwartz & Howard, 1981), the value belief norm theory (Stern, 2000), and the theory of planned behavior (Ajzen & Madden, 1986) researchers (e.g., Bamberg, Hunecke, & Blöbaum, 2007) have shown that perceived social norms in favor of pro-environmental behavior lead to the generation of respective personal norms, which in turn influence pro-environmental

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behavior intentions and behavior itself (for an overview see e.g., [Lindenberg & Steg, 2007](#)). These findings have been complemented by research emphasizing the role of situational defense strategies, such as justifications and excuses for environmentally harmful behavior ([Fritzsche, 2005](#)). This research showed that acceptance ([Schahn, Dinger, & Bohner, 1995](#); [Tyler, Orwin, & Schurer, 1982](#)) and situational availability ([Fritzsche, 2003, 2005](#)) of such defenses determined whether or not people complied with salient pro-environmental norms.

Different norms might often apply to the same situation leading to contradictory predictions for behavior. For instance, in a situation where one needs to choose between a disposable or a reusable cup of coffee at a public dispenser, norms of pro-environmental conduct and also those of efficiency or fashion may apply. That is, even people who accept pro-environmental norms might take the disposable cup if they simultaneously value norms of efficiency. Taking the disposable cup would mean that they would not need to bring the cup back for cleaning. To enhance the predictive value of social norms in situations like these [Cialdini et al. \(1991\)](#) developed the norm focus theory, stating that only those norms which are focused in a given situation predict behavior. Norm focus might either arise from dispositional inclinations to pay attention to certain norms or from situational conditions making a certain norm salient. In support of their assumptions, salience of prescriptive – or “injunctive” – norms (what ought to be done)¹ against littering led to reduced littering in the field. For instance, putting handbill messages such as “April is Keep Arizona Beautiful Month. Please Do Not Litter!” under the windshield wipers of parked cars decreased the likelihood that returning drivers would throw the handbill on the parking lot compared to messages highlighting more distal or different norms ([Cialdini et al., 1991](#)). Littering was also reduced when the returning drivers saw someone picking up litter from the floor, indicating that anti-littering norms apply to the situation ([Kallgren, Reno, & Cialdini, 2000](#); [Reno, Cialdini, & Kallgren, 1993](#); Study 2). [Kallgren et al. \(2000\)](#) also showed that those participants who strongly endorsed anti-littering norms personally littered less when they focused on personal standards (by watching a video of themselves vs. one displaying geometric shapes; Study 3), and when situational norm salience was accompanied by increased arousal (Study 1). In a study by [Goldstein, Cialdini, and Griskevicius \(2008\)](#) descriptive norm messages in hotel rooms increased reuse of towels, particularly when the normative reference group was in the same situation as the participant (“the majority of guests in this room reuse their towels” [italics added by the authors]). In the present work we investigate whether the effects of pro-environmental norm salience are stronger under conditions of cultural worldview defense motivation, which should be increased when people are confronted with existential threat.

1.2. Terror management theory

Terror management theory ([Greenberg et al., 1997](#); [Solomon, Greenberg, & Pyszczynski, 2004](#)), based on the body of work by cultural anthropologist Ernest Becker (1973), proposes that much of human social behavior is rooted in coping with the basic knowledge of human mortality. Anticipation of one's own death is conceived of as separating humans from animals. The combination of a basic striving for self-preservation and resulting fear of death on the one side and at the same time knowing about the inevitability of one's own death on the other is assumed to create the

potential for paralyzing existential terror, disabling adaptive functioning in daily life. Thus humans are proposed to have phylogenetically developed anxiety buffer strategies to prevent existential terror. Both holding a valid cultural worldview and perceiving oneself as living in accordance with the standards of this worldview (i.e., self-esteem) are assumed to buffer existential threat. This is because being a valuable member of one's own culture may allow the self to transcend beyond individual death and thus might provide a sense of symbolic immortality.

To test this theory empirically, [Rosenblatt, Greenberg, Solomon, Pyszczynski, and Lyon \(1989\)](#) investigated the hypothesis that people who are reminded of their individual mortality (e.g., by writing about thoughts and emotions which are associated with their own death vs. writing about dental pain) should be inclined to defend their cultural worldviews and to protect self-esteem. In support for this hypothesis, they found increased punishment intentions against people who deviated from societal rules (e.g., prostitutes) under conditions of mortality salience (MS). In the meantime more than 350 studies have been conducted ([Greenberg, in press](#)) which have shown the effectiveness of MS in increasing self-esteem striving (e.g., by self-serving attributions: [Mikulincer & Florian, 2002](#)) and cultural worldview defense, such as social consensus estimates ([Pyszczynski et al., 1996](#)), liking of stereotypical others ([Schimel et al., 1999](#)), ingroup favoritism ([Castano, Yzerbyt, Paladino, & Sacchi, 2002](#); [Harmon-Jones, Greenberg, Solomon, & Simon, 1996](#)) and defense ([Fritzsche & Jonas, 2005](#)), as well as aggression against dissenting others ([McGregor et al., 1998](#)) and punishment of moral transgressors ([Florian & Mikulincer, 1997](#)). In addition to these rather negative forms of worldview defense, [Jonas, Schimel, Greenberg, and Pyszczynski \(2002\)](#) found increased pro-social behavior towards ingroup members (donations for a domestic charity). Obviously, existential threat has the potential to increase behavior which is consistent with prescriptive norms of a cultural ingroup. This position is also consistent with existential approaches to risk perception which argue that existential threats such as uncertainty or death might be reduced by trusting social groups ([Langford, 2002](#)). The implication is that threat might increase people's inclination to conform to group norms.

However, cultural worldviews often consist of many, very different norms and standards. As such, some of them might contradict each other. For example, Western cultures value both action on behalf of common interests, such as pro-environmental behavior, as well as action serving self-interest. Thus, it is important to ask exactly which norms determine behavior under conditions of existential threat. Recently, [Jonas et al. \(2008\)](#) have addressed this question by combining TMT and norm focus theory ([Cialdini et al., 1991](#)) proposing that the direction of MS effects depends on which social norm is actually salient. They investigated this idea by making specific norms salient, such as pacifism or helpfulness, or opposite norms, such as prosocial vs. proself norms or security vs. benevolence norms of treating crime suspects and found that MS increased behavior and intentions into the direction of the salient norm. This provided the first supportive evidence for a norm focus approach to threat effects on norm relevant behavior. In the present article we apply this approach to the domain of pro-environmental behavior.

1.3. Existential threat and environmental behavior

The role personal threat plays in the prediction of pro-environmental behavior is a longstanding issue within conservation psychology. Here, past research focused mainly on threats *immanent* in the environmental crisis, such as threat to personal wealth or personal health risks in polluted cities (e.g., [Homburg & Stolberg, 2006](#)). These models focus on the specific nature of environmental

¹ [Cialdini et al. \(1991\)](#) distinguish injunctive from descriptive norms where the latter refer to the perceived actual behavior within a culture rather than to cultural prescriptions.

threats, such as severity, likelihood of occurrence, coping potential, and associated causes (Böhm & Pfister, 2005), and predict the effects specific threat appraisals have on environmental risk perception and respective action tendencies.

In the present research, we are interested in the effects of existential threat on pro-environmental action. This kind of threat can – but does not necessarily have to – originate in environmental crisis such as the increasing likelihood of deadly hurricanes. However, it is thought to be ultimately based on the fundamental human problem of being aware of one's own mortality. Although existential thinking has inspired qualitative research on how humans deal with the environmental crisis in order to assuage fears of meaninglessness, uncertainty or death (Langford, 2002), systematic evidence for the function of terror management on pro-environmental behavior is virtually absent in the literature. However, there is some related research on the effects of MS on attraction to wilderness as well as on sustainable behavior in a commons dilemma game. Interestingly, at first glance, both lines of research suggest that existential threat might decrease pro-environmental behavior and attitudes.

Koole and van den Berg (2005) found that wilderness, in contrast to cultivated nature or urban environments, increased the inclination to think about death and that death reminders decreased subjective beauty of wilderness. They argue that due to its association with nature's principles, wilderness reminds people of their mortality and is thus disregarded under conditions of existential threat. We think that these results might have some implications for specific targets of pro-environmental behavior, such as preserving untouched nature, which might appear less desirable under existential threat. However, as pro-environmental behavior is far more than merely preserving wilderness, these results cannot be transferred to environmental protection in general. According to the distinction between different concerns involved in pro-environmental motivation, such as egoistic, altruistic and biospheric concern (Schultz, 2001), pro-environmental behavior might often be motivated by the desire to contribute to the sustained existence of culture. Therefore, pro-environmental motivation might not only be inherently pro-social but also a means to highlight one's cultural existence and to buffer existential anxiety.

Kasser and Sheldon (2000) investigated behavior in an environmental dilemma situation. They found that following MS people became greedy consumers of scarce natural resources when engaging in a forest-management simulation and imagining themselves as owners of a logging company competing with and bidding against three other companies to harvest trees in a national forest (Kasser & Sheldon, 2000, Study 2). In fact, opportunities for pro-environmental behavior often have the nature of a commons dilemma (e.g., Mosler, 1993). Therefore this result may indicate that pro-environmental action is less likely after mortality salience. Alternatively, one may argue in line with a norm focus approach to terror management processes, such that self-interest norms could have been salient in the economic game. Decreased cooperation following MS might reflect behavior in line with this salient social norm. The findings by Kasser and Sheldon would then not say anything about a general tendency of greed or anti-environmental behavior after MS but rather about the influence of norm salience on the direction of MS effects.

In a recent study on environmental concern Vess and Arndt (2008) have provided the first evidence that existential threat can also have beneficial effects on environmental attitudes: For those participants who indicated that their self-esteem was highly contingent on environmental action, MS increased environmental concern in a fictitious case scenario. MS decreased environmental concern, however, in participants who did not derive self-esteem from environmental action. This shows that despite a basic

tendency of people to distance from the natural environment under conditions of existential threat, there might be conditions under which concern for nature contributes to buffer existential fears. As natural environments are often culturally significant and pro-environmental action is often culturally valued, existential threat might lead to increased pro-environmental behavior when pro-environmental norms are salient.

1.4. The present research

The aim of the present research is to investigate the interactive effects of self threat and salience of pro-environmental norms on conservation attitudes and behavior. In contrast to the position that under conditions of self threat people might take less care of the natural environment, we think that salient social norms have a decisive influence on whether perceived threat leads to more or to less pro-environmental behavior. Thoughts related to one's own death might be one of the most basic threats to the self possible, indicating the finite nature of the self. Adopting a terror management framework, we investigate the effects this kind of threat has on pro-environmental conduct. Specifically, we hypothesize that MS increases pro-environmental behavior and related favorable cognitions when pro-environmental norms are focused in a situation. No such effect is predicted when pro-environmental norms are not in focus and a decrease in pro-environmental tendencies following MS is expected when norms are salient that oppose pro-environmental conduct.

In three experimental studies we tested our central assumption that MS increases pro-environmental behavior and cognitions under conditions of focal pro-environmental norms in both the US and Germany. In Study 1 we manipulated MS in a classical fashion by asking our participants to write about emotions associated with death vs. dental pain (control group). As a priming of the pro- or anti-environmental norm, half of the participants witnessed a research assistant who cleaned garbage from the lab, whereas the other half watched the assistant kick the garbage around. As the central dependent variables we measured liking of someone who advocated for an environmentally friendly car and preference for entering a lottery to win a movie on either environmental or neutral issues. Study 2 was a conceptual replication of Study 1 in the context of a forest management game which was adapted from Kasser and Sheldon (2000, Study 2). After a subliminal manipulation of MS, a short text made either a (pro-environmental) common interest norm or a (anti-environmental) self-interest norm salient. Then, we measured participants' intended resource use. In Study 3 we manipulated MS with a subtle priming procedure and compared an explicit pro-environmental norm prime group with participants in a neutral condition. We recorded both intentions to take either a disposable or a reusable cup as well as actual behavior.

2. Study 1

Contemplating environmental issues and holding positive attitudes towards pro-environmental behaviors are basic conditions for pro-environmental behavior. In Study 1 we investigated the interactive impact perceptions of existential threat and salience of pro- or anti-environmental norms have on examination of environmental topics and on approval of pro-environmental consumption. We hypothesized that MS should increase liking of an advertisement for an eco-friendly car;² pro-environmental

² This advertisement article was contrasted with an advertisement for a different car which was not advertised as eco-friendly. This advertisement served as a control stimulus for which no effect of the independent variables was expected.

cognitions, and the relative preference for appropriating a DVD movie on environmental (vs. neutral) topics in the pro-, compared to the anti-environmental norm condition.

2.1. Method

2.1.1. Participants and design

Eighty-five (60 female, 25 male) undergraduates at the University of Rochester with a mean age of $M = 20.02$ years ($SD = 1.12$) participated in this study for extra course credit. The paper-and-pencil study was introduced after the participants had finished an online study and was described as a projective life attitude assessment that assessed general attitudes on public issues. Two participants had to be excluded from analyses because they had guessed the purpose of the study resulting in a final sample of 83 participants.

The study had a 2 (mortality salient; MS/mortality not salient; MNS) $\times$ 2 (pro-/anti-environmental norm salient) design with approval of environmentally friendly consumption and examination of environmental topics as dependent variables.

2.1.2. Procedure

Participants were told that the study consisted of two unrelated parts. The first part was an online study which was accessed on a computer in the laboratory ingroup sessions with between 3 and 5 participants per session. It was described as a problem-solving task that included the measurement of performance. The second part, a paper-and-pencil survey, assessed the participants' pro-environmental attitudes. When they had finished the survey, they were told that they could choose to participate in a lottery, in which they could win either a movie on ecological topics or a crime movie not related to environmental topics. On the ticket that they drew, the participants read an email address to which they were told to send their respective survey ID in order to participate in the lottery. The ID served to identify the condition of the participant while holding the identity of the participant anonymous.

2.1.2.1. Norm focus manipulation. The purpose of the first part, which took about 15 min, was to manipulate the salience of pro-environmental norms. We were interested in the effect of a *generalized* environmental norm rather than the effects of proximal norms concerning *specific* behavior. Therefore, the domain in which the pro-environment norm was made salient (littering) differed from the domains of outcome behavior (ecological consumer behavior; search for environmental information).

Right before the online study began, a female experimenter who coordinated the entire experiment spread garbage in the laboratory. In the *pro-environmental norm* condition, at the end of the online study, the experimenter apologized for the condition of the lab and ostentatiously started to clean it up. The cleaning procedure lasted at least 5 min and continued after the participants were introduced to the apparent follow-up study. In the *anti-environmental norm* condition, the experimenter kicked the garbage around and remained ignorant toward comments if participants criticized the bad condition of the lab. She continued kicking the garbage during the end of the online study and at the beginning of the follow-up.

2.1.2.2. Mortality salience manipulation. The paper-and-pencil questionnaire started with an explicit MS manipulation similar to the standard procedure used in prior terror management research (Greenberg et al., 1997). In the *MS condition* participants were asked to describe the emotions and the thoughts that their own death arouses in them. In the *MNS condition*, the participants should describe the emotions and the thoughts that their last dentist visit

aroused in them. The following page contained two filler questionnaires, namely a mood questionnaire (PANAS; Watson, Clark, & Tellegen, 1988)³ and a word puzzle.

2.1.2.3. Dependent variables. Participants read two different advertisements for cars. One described a SUV (sport utility vehicle), namely the FORD Expedition, and the other described a hybrid car, namely the Toyota Prius. Both articles were created for the purpose of this study and tested in a pilot study for credibility. The order of presentation of the two articles was counterbalanced.⁴ To assess approval of environmentally friendly consumption, participants were asked to indicate their liking of the article on ten-point Likert-type scales from 1 (not at all) to 10 (very much) for each article. For each article they were asked the following items: "How much do you like the person who wrote this article?"; "How intelligent do you think this person is?"; "How well-informed do you think this person is?" (for Ford Expedition, $\alpha = .90$; for Toyota Prius, $\alpha = .72$).

At the end of the study, we measured participants' actual information search behavior with regard to examination of environmental topics. After the participants finished completing the survey, they were invited to participate in a lottery. They were told that by participating they could win a DVD of one of two films, which they could choose. The movie on environmental issues was "The Swarm", a thriller described to involve the sea, underwater world and ecological issues in Cape Cod. Alternatively the participants could choose "Swimming-Pool", a thriller described to involve a beach vacation in Fort Lauderdale. If they were interested, they could choose to draw a lottery-ticket for either the Swarm lottery or for the Swimming-Pool lottery. Both displayed the respective email address for either lottery. They were asked to contact this email address along with their participant ID (for later condition recognition). If they were not interested in participating, their decision counted as no DVD choice condition.

After the experiment the participants were debriefed about the purpose of the first part and thanked for their participation. To ensure the assessment of the participants' preference to examine environmental topics, only after a two-week period were they debriefed about the purpose of the second part of the study.

2.2. Results

2.2.1. Approval of environmentally friendly consumption

We submitted the liking scores for the car advertisement article to a 2 (MS/MNS) $\times$ 2 (pro-/anti-environmental norm) $\times$ 2 (evaluation of Expedition/Prius) ANOVA with repeated measurement on the last factor (for cell means see Table 1). We found a main effect of car indicating that overall, people liked the Prius advertisement more than the Expedition advertisement, $F(1, 79) = 6.28$; $p = .01$. Furthermore, we received two-way interactions of car and norm prime, $F(1, 79) = 7.99$; $p = .006$, and of car and MS, $F(1, 79) = 2.80$; $p = .098$, marginal. These interactions were qualified by the predicted three-way interaction of MS, norm prime, and car, $F(1, 79) = 5.25$; $p = .025$. Simple effects analyses⁵ showed that MS increased liking of the Prius advertisement only in the pro- but not in the anti-environment norm condition, $F(1, 79) = 7.76$; $p = .007$. Looked at differently, the pro- (vs. anti-) environment norm prime increased the liking of the Prius advertisement only when mortality

³ The experimental manipulations did not affect mood ratings and were thus not included in further analyses. This finding is in line with prior research on MS effects (e.g., Greenberg et al., 1997).

⁴ As order of presentation did not affect our results, we do not consider this variable in the analyses.

⁵ We applied the Bonferroni adjustment procedure to all simple effect analyses in this article.

Table 1

Attitudes towards pro-environmental and neutral car and choice of environment related movie as a function of mortality salience and norm salience in Study 1.

	Mortality salient		Mortality not salient	
	Pro-environmental norms salient	Anti-environmental norms salient	Pro-environmental norms salient	Anti-environmental norms salient
N	20	22	22	19
Liking Prius advertisement	$M = 7.40$ $SE = .27$	$M = 5.96$ $SE = .25$	$M = 6.38$ $SE = .25$	$M = 6.11$ $SE = .27$
Liking expedition advertisement	$M = 5.75$ $SE = .33$	$M = 6.12$ $SE = .31$	$M = 6.14$ $SE = .31$	$M = 6.05$ $SE = .34$
Selection of environment related DVD	$M = .35$ $SE = .12$	$M = -.32$ $SE = .11$	$M = .14$ $SE = .11$	$M = -.11$ $SE = .12$

was salient, $F(1, 79) = 15.56$; $p < .001$. Finally, the Prius was preferred over the Expedition advertisement only when pro-environment norm and mortality were salient, $F(1, 79) = 20.91$; $p < .001$. No other simple effects reached significance, all F s < 1 .

2.2.2. Information search behavior

We recoded choice of DVD such that participants received a value of 1 if they had chosen the environment related movie and a value of -1 if they had chosen the movie not related to environmental issues. If they had chosen no movie at all we assigned a value of 0. We submitted this index of choosing the environment related DVD to a 2 (MS/MNS) $\times$ 2 (pro-/anti-environmental norm) ANOVA, revealing a main effect of norm condition, $F(1, 79) = 15.57$; $p < .001$ which was qualified by the predicted interaction of norm condition and MS, $F(1, 79) = 3.42$; $p = .068$ (marginally significant). According to simple effect analyses, the salience of a pro-environmental norm only led to increased choice of the environment relevant DVD when mortality was salient but not when mortality was not salient, $F(1, 79) = 17.03$; $p < .001$. No other simple effects occurred. For cell means see Table 1.

2.3. Discussion

The results of Study 1 support our assumption that existential threat can lead to increased pro-environmental attitudes when pro-environmental norms are in focus. At the same time, mortality salience moderated the impact of salient pro-environmental norms on approval of environmentally friendly consumption and the examination of environmental issues.

It is noteworthy that the manipulation of littering acceptance affected behavior which was not directly related to the act of littering. Instead, participants obviously generalized from the (anti-)littering norm to a global (pro-) environmental norm. However, given the findings of Cialdini et al. (1991), who showed that the impact of injunctive norm messages on behavior decreased with decreasing relatedness between the content of the salient norm and behavior, the present effects might reflect a rather conservative estimation of normative impact under conditions of existential threat. However, as pro-environmental norms are often formulated at a rather abstract level, the result might come close to what we would expect to happen under natural conditions. Furthermore, medium relatedness between norm prime and attitude and behavior content might be an explanation for the absence of any effect of norms on behavior in the MNS condition. In the MS condition, one might speculate that participants were highly vigilant for cues in the environment which would indicate the appropriate norms in that situation.

In contrast to what we expected, MS did not lead to less pro-environmental cognitions in the anti-norm condition. This might

be due to failure to induce an anti-environment norm in Study 1. Kicking around garbage might not have been perceived as an expression of a respective social norm but perhaps was attributed to carelessness of the experimenter or to a lab-specific norm which did not generalize to the world outside the lab.

Although approval of less polluting cars and a basic interest in environmental issues can be important pre-conditions for behavior consistent with pro-environmental social norms, Study 1 did not address actual pro-environmental behavior. This scope is broadened in Study 2 which assessed behavioral intentions in an environmental dilemma.

3. Study 2

In Study 2 we aimed to conceptually replicate the findings of Study 1 for behavioral intentions in an environmental commons dilemma. In the commons dilemma game, participants make decisions which either serve to maximize individual profit or enhance sustainable cooperative management of a public good, such as the natural resources fresh water or clean air. Thus it has often been used to model and measure pro-environmental (i.e. resource preserving) behavior (e.g., Kortenkamp & Moore, 2001; Mosler, 1993). Furthermore, we intended to take norm focus theory's assertion serious that sometimes *contradictory* social norms can potentially be applied to a specific situation and that focus then determines which of them will influence behavior in the respective setting. Thus, we made two different norms salient which suggested contradictory behaviors rather than only activating a relevant norm and comparing this to a neutral condition. In order to do this we borrowed the forest-management simulation paradigm from Kasser and Sheldon (2000, Study 2) and incorporated either a self-interest norm prime or a common interest norm prime into the materials.⁶ In order to increase generalizability of these effects and increase our confidence that death-related cognition is the key variable, we did not manipulate mortality salience as in Study 1. Instead we used subliminal death primes to manipulate the accessibility of death-related thought, employing a procedure based on Arndt, Greenberg, Pyszczynski, and Solomon (1997).

As indicated before, we think the finding by Kasser and Sheldon shows that people become greedier following mortality salience when competitive values are salient. The specific language in the Kasser and Sheldon materials indicated that the goal for participants was to cut as many acres of trees as they could in order to maximize profit for their company. If this made the value or norm of out-competing others salient, then our norm focus and TMT

⁶ Please note, that self-interest and common interest resemble self-enhancement and self-transcendence values proposed by Schwartz (1992). As values represent cultural ideals, and reflect normative hierarchies within cultures people's priorities to the values will be to a large extent determined by cultural norms.

integration qualifies their finding. In order to test our hypothesis, we removed the language indicating that the participants' goal was to maximize the profit of their company, and added an introduction to the forest-management simulation paradigm in which we either primed a common interest norm or a self-interest norm. The common interest norm was primed by pointing out the problems of common resources and the danger of ruining them if people strive to maximize their individual profit. The self-interest norm was primed by pointing out the benefits of the market economy in which all people strive to maximize profits. We predicted that for people in the common interest norm conditions, a death prime would lead people to behave less greedily than a control prime, whereas the reverse pattern, replicating [Kasser and Sheldon \(2000\)](#), would occur for people in the self-interest norm conditions.

3.1. Method

3.1.1. Participants and design

One hundred and four introductory psychology students (55 women and 49 men) from the University of Arizona participated in this experiment for partial course credit. Participants were randomly assigned to one of four conditions in a 2 (subliminal prime: dead vs. pain) $\times$ 2 (norm: common interest vs. self-interest) between-subjects design. Three participants were excluded from the analysis because they did not follow the instructions on the primary dependent measure.

3.1.2. Procedure

Upon arriving for the experimental session, the experimenter informed the participants that they would take part in two brief and unrelated studies. The first study was described as an investigation of the effects of distraction on word identification that participants would complete on a computer. This task served as the mortality salience manipulation. The second study was described as a study of decision-making processes in which participants would be asked to fill out a questionnaire packet. The decision-making questionnaire contained the norm manipulation and the dependent measures. All participants completed each part of the session in individual rooms equipped with a computer.

3.1.2.1. Mortality salience manipulation. After giving an overview of the session, the experimenter introduced the word identification task, which was a modified lexical decision task. For this task participants first saw a letter string (a "distracter") flashed briefly on the screen followed by a second letter string that was either a word or non-word. Participants were instructed to identify whether the second letter string was a word or a non-word as quickly as possible using the keyboard. The distracter preceding the target word consisted of a forward mask, a prime word and a backward mask. The letter strings "YLDQFBY" and "QHYPDQ" served as the forward and backward masks. The forward and backward masks were presented for 167 ms each. In the death prime condition, the word "DEAD" was presented for 33.4 ms (two frames at 60 Hz refresh rate) between the two letter string masks. In the control condition, the word "PAIN" was presented between the masks ([Arndt et al., 1997](#)). Participants controlled the presentation of each subsequent set of letter strings, which appeared 1 s after the participant provided a response to the lexical decision task. The entire task consisted of five practice trials and 30 priming trials.

3.1.2.2. Norm focus manipulation. After participants completed the lexical decision task, the experimenter handed out the questionnaires for the "study on decision-making". Norm activation (common interest vs. self-interest) was accomplished by presenting participants with "general information" at the beginning of the

questionnaire. For participants in the common interest condition this general information described the destruction of valuable natural resources in what was essentially an illustration of [Hardin's \(1968\)](#) "tragedy of the commons". The text outlined the conflict between individual and group interest in short-term and long-term resource utilization strategies for managing natural resources. Participants in the self-interest norm condition read general information about the market economy and the idea of the "invisible hand" first described by [Adam Smith \(1776\)](#). The text outlined why people should behave in their own interests by explaining some of the economic benefits that are created when people pursue their personal interests and how in the end all people benefit from the wide range of products that result from competition.

3.1.2.3. Forest management game and dependent variables. After reading the introductory information, participants received the specific decision problem, which was adapted from [Kasser and Sheldon \(2000, Study 2\)](#). In this procedure participants are asked to imagine that they were the owner of a timber company. Their company and three other timber companies were all working within the same national forest containing 2000 acres of timbered land. Each year, the four companies bid on how many acres they would cut for that year, with a maximum bid of 100 acres. None of the companies were aware of what the other companies' bid; they only knew the total number of acres cut by the four companies each year. This total amount could range from 0 (if all 4 companies bid 0) to 400 (if all 4 companies bid 100). Thus, in a given year, the forest could be reduced by as much as 400 acres (i.e., in year one, the forest could shrink from 2000 to 1600 acres). Participants were also informed that the forest regenerated at a rate of about 10% per year. Furthermore, it was pointed out that one danger was that the forest would eventually be wiped out, leaving all 4 companies "out in the cold." This point implied that it was to the collective advantage of the four companies to make small yearly bids. However, the point was also made that a company would not do as well financially if it cuts less than the other three companies. Thus, it would be to each company's individual advantage to make larger bids. As mentioned earlier, we removed language indicating that the participants' goal was to cut as many acres as they could in order to maximize the profit of their company, which we believed would have made salient the value of out-competing others across all the conditions.

Following the introductory information, participants were asked to answer several questions based on how *they* would think and behave in this situation, namely: (1) To what extent would you expect the other companies to cut small amounts each year? (2) To what extent would you expect the other companies to cut maximum amounts each year? (3) To what extent would you prefer to profit more than the other three companies? (4) To what extent would you prefer every company to have similar levels of profit? (5) To what extent do you expect the other companies to cut excessive amounts? (6) To what extent would you like to cut more than the other companies? Responses were made on a scale from 1 = *not at all* to 7 = *very much*.

Then they were asked to indicate how many acres they would cut in the bid for the first year (ranging from 0 to 100 acres). Next, they were asked to respond to six reasons as to why they had selected their bid: (1) because it was the most fair or ethical bid to make, (2) because it was the best strategy for competing against the other companies, (3) because it was the most morally correct bid to make, (4) because they would hope to establish a sense of cooperation among the companies, (5) in order to make more profit than the other companies, (6) so that all the companies could do well. Participants were instructed to indicate to what extent each reason applied to them on a scale from 1 (*not at all*) to 5 (*very much*).

Then participants were asked to imagine they had discovered, after having made their first bid, that they and the other three companies had cut 300 acres total (out of a maximum of 400) in the first year. The question they responded to was, “How many acres would you cut in the *second* year?” (ranging from 0 to 100 acres). Then they were again asked why, choosing from the six reasons outlined above. Finally, they were asked to imagine another alternative that they and the other three companies had cut only 100 acres total (out of a maximum of 400) in the first year. They were then asked how many acres they would cut in the *second* year in this instance (ranging from 0 to 100 acres) and again, they chose one of the six reasons offered for why they made that bid.

After participants finished the decision task they completed another questionnaire that contained questions about the word identification task (dead vs. pain priming), in which we presented participants four words (*boat*, *dead*, *lump*, and *pain*) and asked them to select the word they thought they might have seen in the word identification task. Then participants were debriefed and thanked for their participation.

3.2. Results

3.2.1. Masked prime awareness check

Whereas 53 participants checked the word “lump” and 28 “boat”, only 9 participants thought they had seen the word “dead” (2 in the dead and 7 in the pain condition) and only 11 thought they had seen “pain” (7 in the dead and 4 in the pain condition). This finding indicates that participants were not aware of the masked subliminal prime.

3.2.2. Gender

We next checked for gender effects on the dependent variables and ran a 2 (gender) $\times$ 2 (subliminal prime: death vs. pain) $\times$ 2 (norm: common interest vs. self-interest) $\times$ 3 (bid: bid 1 vs. bid 2 vs. bid 3) ANOVA with repeated measures on the fourth factor. Although we found that male participants overall wanted to cut more acres ($M = 64.15$, $SD = 18.89$) than females ($M = 54.52$, $SD = 21.73$), $F(1, 93) = 5.41$, $p < .03$, there was no interaction of gender with the experimental conditions (all $ps > .12$). Although a chi-square test indicated participants' gender was equally distributed across experimental conditions, in the analysis reported below we considered gender as additional factor.

3.2.3. Bids

The 2 (subliminal prime: death vs. pain) $\times$ 2 (norm: common interest vs. self-interest) $\times$ 2 (gender) $\times$ 3 (bid 1 vs. bid 2 vs. bid 3) ANOVA with repeated measures on the fourth factor yielded, in addition to the above reported gender effect, a significant two-way interaction between the subliminal death vs. pain prime and the three bids, $F(1, 93) = 7.09$, $p < .01$, which was qualified by a significant three-way interaction with the norm factor, $F(1, 93) = 4.49$, $p < .04$.

Next we conducted separate 2 (subliminal prime: death vs. pain) $\times$ 2 (norm: common interest vs. self-interest) ANOVAs for each of the three bids. For the first year we found a main effect for the subliminal prime, $F(1, 97) = 7.39$, $p < .01$, with participants in the death prime condition wanting to cut fewer acres ($M = 50.05$, $SD = 27.79$) than participants in the pain prime condition ($M = 63.70$, $SD = 21.57$). For the second bid (i.e. participants were informed that in the first year the companies altogether had cut 300 acres) the results revealed no significant effects, all $ps > .11$ (with a grand mean of $M = 65.01$, $SD = 26.05$). For the third bid (i.e. participants were informed that in the first year the companies had cut 100 acres altogether) the results showed the predicted two-way interaction, $F(1, 97) = 5.48$, $p < .03$. The means for the third bid are

presented in Table 2. Simple effects analysis revealed that whereas there was no difference between the two norms in the pain condition, $t(97) = 1.20$, $p > .23$, participants in the death condition focusing on the common interest norm made lower bids than participants focusing on the self-interest norm, $t(97) = 2.12$, $p < .04$. Simple effects within each level of the norms revealed that in the common interest norm condition, participants primed with death tended to reduce their bids compared to the pain condition, $t(97) = 1.66$, $p = .10$, whereas for the self-interest norm, participants in the death condition tended to increase their bids, $t(97) = 1.65$, $p = .10$.

3.2.4. Expectations of other companies and preferences for own behavior

A factor analysis using the principal component analysis with the ratings on the first six questions regarding participants' expectations and preferences yielded two factors with eigenvalues greater than 1 ($KMO = .62$): “bid expectations” (e.g. To what extent do you expect the other companies to cut excessive amounts?) (eigenvalue 2.17, explained variance 36.23%, Item 1 reversed, Item 2, Item 5, factor loadings .77–.79, Cronbachs alpha = .68), and “preferences to profit” (e.g. To what extent would you prefer to profit more than the other three companies?) (eigenvalue 1.58, explained variance 26.37%, Item 3, Item 4 reversed, Item 6, factor loadings .68–.84, Cronbachs alpha = .70). Conducting two separate 2 (subliminal prime: death vs. pain) $\times$ 2 (norm: common interest vs. self-interest) ANOVAs for the two factors (bid expectations and preferences to profit) revealed that in the death condition participants expected the other companies to cut down fewer trees each year, $M = 4.31$, $SD = 1.32$, than in the pain condition, $M = 4.98$, $SD = 1.16$, $F(1, 97) = 7.27$, $p < .01$. There were no effects regarding participants' preferences to profit, all $Fs < 1.2$.

3.2.5. Reasons for choices

We conducted a factor analysis using the principal component analysis with the 18 items regarding participants' reasons for their choices and found four factors ($KMO = .83$), “moral” (e.g. Because it is the most fair or ethical bid to make.) (eigenvalue 7.48, explained variance 41.55%, Items 1 and 3 for the three bids, factor loadings .73–.79, Cronbachs alpha = .90), “cooperation” (e.g. Because I would hope to establish a sense of cooperation among the companies) (eigenvalue 2.48, explained variance 13.80%, Items 4 and 6 for the three bids, factor loadings .71–.82, Cronbachs alpha = .89), “competition regarding bid 1 and bid 3” (e.g. In order to make more profit than the other companies,) (eigenvalue 1.86, explained variance 10.32%, Items 2 and 5 of first and third bid, factor loadings .72–.82, Cronbachs alpha = .87) and “competition regarding bid 2” (e.g. In order to make more profit than the other companies,) (eigenvalue 1.44, explained variance 7.97%, Items 2 and 5 of second bid, factor loadings .72–.80, Cronbachs alpha = .86). We submitted each factor (moral, cooperation, competition regarding bid 1 and bid 3, competition regarding bid 2) to a 2 $\times$ 2 ANOVA and found the predicted interaction for the factor “competition regarding bid 1 and bid 3”, which paralleled the interaction for the

Table 2

Acres of wood intended to cut for the third bid (2nd year/100 acres in 1st year) as a function of mortality salience and norm salience in Study 2.

	Mortality salient		Mortality not salient	
	Common interest norm	Self-interest norm	Common interest norm	Self-interest norm
<i>N</i>	26	25	24	26
<i>M</i>	47.69	63.48	60.21	51.15
<i>SE</i>	27.21	30.79	28.83	18.46

third bid, $F(1, 97) = 6.32, p < .02$. Whereas there was no significant difference between participants' responses in the pain condition following the common ($M = 3.41, SD = 1.05$) or self-interest norm activation ($M = 3.20, SD = 1.14$), $t(97) < 1, p > .50$, participants were less inclined to make more profit than the other companies in the death condition following the common interest norm activation ($M = 2.83, SD = 1.13$), compared with the self-interest norm activation ($M = 3.71, SD = 1.01$), $t(97) = 2.90, p < .006$. Looked at differently, with the activation of the common interest norm, the death reminder tended to decrease participants' inclination to compete, $t(97) = 1.88, p < .07$, whereas with the activation of the self-interest norm, the death reminder tended to increase their inclination to compete, $t(97) = 1.70, p < .10$. No significant effects were found for the remaining three factors (all $F_s < 2$).

3.3. Discussion

The results of Study 2 showed that mortality salience increased conservationist behavior prior to any information about the other companies' behavior (see results for bid 1). Although there were no significant effects of mortality salience on participants' bids when they imagined that the other companies cut down three quarters of the trees in the first year (bid 2), we did find the predicted interaction when participants imagined that the other companies only cut down a small portion of trees in the first year (see results for bid 3). In this case, mortality salience led to less consumption in the common interest norm condition and more consumption in the self-interest norm condition. Though the results for the second bid might appear, on the surface, to speak against the interactive nature of mortality salience and norms, a closer examination of this bidding condition suggests otherwise. Knowing that the other companies together cut all but 100 acres could have directed or constrained participants' bids by communicating a strong norm of consumption. Indeed, Cialdini and colleagues have manipulated norm focus in a similar manner by simply allowing people to observe the actions of another person (Kallgren et al., 2000). The constraints on participants in bid 2 may have been further compounded by placing a significant limit on the number of acres participants had to work with. Thus, participants could have felt that both psychologically and numerically their range of responses were limited on the second bid. Compared to bid 3, participants could have felt as if greed was essentially their only choice. Consistent with this explanation, the grand mean for bid 2 ($M = 65.01$) was comparable in size to the mean of the self-interest norm, subliminal death prime condition of bid 3 ($M = 63.48$). In contrast to bid 2, the predicted interaction pattern derived from the combined terror management and norm focus approach did emerge for the third bid when there was more psychological and numerical room (i.e. 300 acres) for participants to feel like they could be either greedy or conservationist.⁷

Our failure to replicate Kasser and Sheldon's (2000, Study 2) finding that mortality salience increased greed overall might stem from our elimination of the statement explaining that the goal of the experiment was to maximize one's own profit. It is also possible that this difference reflected stronger conservationist norms in our Arizona sample than in the Midwestern sample utilized by Kasser and Sheldon. Consistent with this possibility, for the first bid, mortality salience generally decreased people's intentions to

harvest in the first year when there was no information about the behavior of the other companies. In addition, personal norms may help explain differences in our results compared to findings by Dechesne, Pyszczynski, and Arndt (2003). They found that men increased greedy behavior following mortality salience and the effect was in the opposite direction for women. Perhaps our participants, students in the southwest, were generally more environmentally conscious than participants in the previous studies; when they had no clear information about how the other companies would behave, they generally wanted to save natural resources following MS. This effect was paralleled by decreased expectations regarding the other companies' harvesting intentions following MS.

Although we cannot be sure why there was a general tendency toward conservationism under existential threat in this sample, the results of Study 2 do provide further support for our reasoning that norm focus should determine whether MS increases or decreases pro-environmental behavior. When the other companies intended to harvest only 100 acres total, participants in the mortality salience condition reacted in accordance with the salient pro-environmental, common interest norm by making lower bids. This conceptually replicates the findings of Study 1. Extending the results of Study 1, the death prime led people to harvest more (compared to the control condition) when the anti-environmental self-interest norm was activated. In addition, we found the same interaction pattern for participants' inclination to profit more than the other companies. These results suggest that following self-interest norm activation, the death prime increased people's inclination to out-compete others, whereas following the pro-environmental common interest norm, the death prime decreased intentions to out-compete others on the cost of depleting a natural resource.

4. Study 3

The two previous studies showed the interactive effect of existential threat and salience of a pro-environmental norm on information search and pro-environmental attitudes in Study 1 and on conservation intentions in an environmental commons dilemma game in Study 2. In Study 3, we aimed at replicating this basic pattern for actual pro-environmental behavior (using reusable instead of disposable cups). In contrast to Studies 1 and 2 we did not include an anti-environment norm condition but compared a pro-environmental norm salience condition with a neutral control condition. Furthermore, in Study 3 norm salience was manipulated in an explicit manner whereas MS was manipulated subtly.

4.1. Method

4.1.1. Participants and design

We asked 61 male and 66 female students at the University of Jena with an average age of $M = 23.3$ ($SD = 4.5$) to participate in a 15 min study located in a nearby seminar room on campus. We had to exclude 12 participants because they did not take a hot beverage and thus did not provide data for our non-reactive dependent variable. Additionally, we excluded three participants who observed other participants' choices, another three participants due to missing data on questionnaire items, one participant who did not receive the experimental treatment, and one who indicated never using plastic cups in principle. The final sample consisted of 48 men and 59 women.

The study used a 2×2 between-subjects design with the factors MS (MS vs. MNS) and norm salience (environmental norm salient vs. environmental norm not salient).

⁷ As it has been done in the original study by Kasser and Sheldon (2000, Study 2) we always presented the scenarios in the same order. This allowed for a direct comparison between our results and those of the previous study. However, although we do not think that this specific order had an impact on the pattern of effects we cannot fully exclude this possibility.

4.1.2. Procedure and materials

When recruited, participants were told that the study is about marketing and more specifically, about the preferences for different hot beverages. When participants entered the seminar room they were seated at one of four desks each equipped with a notebook computer. All participants faced the front wall of the lecture room with a blackboard on it. First, participants received a brief paper-and-pencil questionnaire on socio-demographic characteristics. Then the computer program started with a short introduction, inviting the participants to take part in a marketing study on hot beverages. Furthermore, the introduction stated that participants would receive a hot beverage at the end of the study and that they would be able to choose between a disposable cup and a reusable cup. They were told that only the reusable cup would have to be given back to the experimenters. Participants then were told to direct their attention to an eight-digit code on the blackboard. The code had to be entered into the computer and was written on the blackboard to assist with a subtle MS induction.

4.1.2.1. Mortality salience manipulation. In the MS condition the blackboard displayed not only the code but also a hand-written stanza of the poem “Eine Leichenphantasie” (A Funeral Fantasy) by Friedrich Schiller, apparently left from the last seminar. Several death-related words were circled in red, namely, “Leichen” (corpses), “erstorben” (died), “trauern” (mourning), “Gruft” (sepulcher), “toten” (dead), “Toten[pompe]” ([pomp of] death) and “Grab” (grave). In the MNS control condition the stanza was slightly changed by replacing all death-related words by neutral ones. The following words were circled: “Mitternachts[phantasie]” (midnight [phantasy]), “hellen” (light), “klaren” (clear), “stillen” (quiet), “blinken” (gleaming), “Duft” (flavor), “Blüten[pompe]” ([pomp of] flower), “Elfen” (elves) and “Mitt[nacht]” (mid[night]).

4.1.2.2. Norm salience manipulation. After participants had entered the code from the blackboard, the computer program continued. Participants had to answer some questions about their preferences for different hot beverages in order to enhance the credibility of the marketing cover story. Subsequently, participants were instructed to watch carefully and later evaluate an advertisement about coffee, which was ostensibly produced by a former participant. The lay advert was displayed in the middle of the computer screen and was roughly 90 s long. The advertisement was used to manipulate the salience of the environmental norm. In both conditions (environmental norm salient and environmental norm not salient) participants saw a confederate (a 22 year old woman) who stood in front of a table. A coffee and a tee pot, and both disposable and reusable cups were placed on the table. Since the editing of the advertisement was apparently not yet done, participants observed the confederate talking casually to the experimenter who stood behind the camera. In the pro-environment norm salient condition, the confederate poured herself a coffee and said “By the way, I appreciate that you’re offering reusable cups and not only such polluting plastic stuff”. The experimenter replied: “Definitely! However, please bring the cup back if you take one of the reusable cups.” In the pro-environment norm not salient condition, the confederate said instead “By the way, I find it rather nice that you have such applied project seminars in your studies. Unfortunately, they are held very rarely at University.” Here the experimenter answered: “Definitely! However, such applied project seminars are indeed held very rarely at University.” In both conditions the lay advert continued with the same advertising slogan. The confederate spoke several sentences directly into the camera about the new *Vigilanz* coffee and its special portion of caffeine. After the confederate finished the slogan, she paused and added somewhat jocosely “*Vigilanz* coffee tastes even better in environmentally friendly

cups” in the pro-environmental norm salient condition or “*Vigilanz* coffee tastes even better in project seminars” in the pro-environmental norm not salient condition. Afterwards the experimenter thanked the confederate and turned off the camera. The advertisement was followed by several questions concerning the quality of the advert. Again, these questions served as filler items to enhance the plausibility of the marketing study.

4.1.2.3. Dependent variables. When the computer program ended, the experimenter told the participant that the study was almost over and that he/she could get him-/herself a hot beverage before a second experimenter would ask some final questions. Participants were led to the end of the room where they encountered a table with a coffee and a tee pot, and both disposable and reusable cups on it. A sign indicated that the reusable cups had to be given back to the experimenter. The experimenter took care that at each time there was only one participant getting him-/herself a hot beverage so that other participants were not influenced by the choice of their predecessors. The second experimenter recorded which cup the participant chose as our central dependent measure. After participants had taken their hot beverage, they filled out the final questionnaire behind a wall so that other participants could not see whether their predecessors had chosen a disposable or a reusable cup. First, participants were asked how satisfied they are with their drink. Then two items measured their prior intentions for both choosing a reusable cup and choosing a disposable cup, independently. All items used Likert-type items with 10-point scales from 1 (“not at all”) to 10 (“very much”). At the end of the study, participants were debriefed and thanked for their participation.

4.2. Results

4.2.1. Behavioral intentions

We calculated a difference measure of behavioral intention by subtracting the inclination values for the disposable from the inclination values for the reusable cup. As exploratory analyses revealed that intention to take a reusable cup was affected by whether people chose coffee ($n = 66$) or tea ($n = 41$), we entered this variable as a factor in our analyses.⁸ Hence, we submitted the difference measure of intention to a 2 (MS/MNS) $\times 2$ (pro-environmental norm salient/not salient) $\times 2$ (coffee/tea chosen) ANOVA (for means see Table 3) revealing a main effect of norm salience, $F(1,99) = 5.27$; $p = .02$, indicating increased intention to take the reusable cup when the pro-environmental norm was salient ($M = .90$; $SD = 6.74$) compared to when the pro-environmental norm was not salient ($M = -2.64$; $SD = 6.29$). A further main effect of kind of drink chosen, $F(1,99) = 4.46$; $p = .04$, indicated that those participants who had chosen coffee were more inclined to take the reusable cup ($M = .30$; $SD = 6.63$) than those who had chosen tea ($M = -2.46$; $SD = 6.65$). Most importantly, however, there was the predicted interaction of MS and norm salience, $F(1,99) = 6.25$; $p = .014$. Simple comparisons show that MS increased intentions to take the reusable cup only when the pro-environmental norm was salient, $F(1,99) = 5.48$; $p = .02$, but not when it was not salient, $F(1,99) = 1.45$; $p = .23$. Looked at differently, pro-environmental norm salience only increased the intention to take the reusable cup when mortality was salient, $F(1,99) = 13.03$; $p < .001$ but not when it was not salient, $F(1,99) < 1$.

⁸ In each cell of the design there were at least 8 participants. Removing kind of drink chosen as a factor from the analyses did not change the results for intention, but for actual behaviour, the interaction of MS and norm salience turned out to be non-significant.

4.2.2. Behavior

For actual behavior decisions (0 = disposable cup; 1 = reusable cup) we again calculated a 2 (MS/MNS) $\times$ 2 (pro-environmental norm salient/not salient) $\times$ 2 (coffee/tea chosen) ANOVA (for means see Table 3) which revealed a main effect for norm salience, $F(1,99) = 6.22$; $p = .014$, indicating increased likelihood of taking a reusable cup when the norm was salient ($M = .49$; $SD = .50$) then when it was not ($M = .22$; $SD = .42$). One further – marginally significant – main effect occurred for kind of drink chosen, $F(1,99) = 3.76$; $p = .055$, showing that participants who had chosen coffee more often took a reusable cup ($M = .44$; $SD = .50$) than those who had chosen tea ($M = .24$; $SD = .44$). Most interestingly and in line with our predictions, the only other effect we found was a marginally significant interaction of MS and norm salience, $F(1,99) = 2.82$; $p = .097$. Simple effect analyses show that when the pro-environmental norm was salient, $F(1,99) = 4.12$; $p = .045$, but not when it was not, $F(1,99) < 1$, MS increased the proportion of reusable cups chosen (MS: $M = .59$; $SD = .50$; MNS: $M = .39$; $SD = .50$). Looked at differently, in the MS, $F(1,99) = 9.86$; $p = .002$, but not in the MNS condition, $F(1,99) < 1$, norm salience increased the choice of reusable cups (norm salient: $M = .59$; $SD = .50$; norm not salient: $M = .21$; $SD = .41$).

4.3. Discussion

In Study 3 we replicated the interaction of MS and pro-environment norm salience found in Studies 1 and 2: MS increased pro-environmental behavior intentions and actual consumption of reusable cups when pro-environment norms were focal and pro-environment norm salience only increased intentions and behavior when existential threat was salient at the same time.

Most notably, in Study 3 we were able to show the predicted pattern for actual pro-environmental behavior which considerably extends the previous studies. This also enhances our confidence in the robustness of our findings and in the proposed notion that individual threat can increase people's pro-environmental behavior under conditions of focal pro-environmental norms.

5. General discussion

Perceptions of individual threat can foster pro-environmental attitudes and behavior under conditions of salient pro-environmental social norms. In addition, reminding threatened people of pro-environmental norms might result in particularly strong responses in the normative direction. We demonstrated this in three independent studies. These studies were conducted in different geographic and cultural contexts and used different experimental paradigms and measures. In Study 1, explicitly manipulated MS and a norm model manipulation interactively determined both pro-environmental attitudes in judging car advertisements and in examining environmental issues. In Study 2, behavioral intentions in an environmental commons dilemma game were determined by the interaction of subliminally

manipulated MS and the presence of norm cues in the written introductions. Finally, in Study 3 the interaction of MS and pro-environment norm salience which were both manipulated subtly, predicted both intention and actual behavior with regard to choice of reusable vs. disposable cups. The variety of different samples, manipulations and dependent variables used in these studies indicates the robust nature of the revealed pattern and increases our trust in its external validity.

5.1. Terror management processes and pro-environmental behavior

The present work is the first that explicitly relates terror management processes to pro-environmental behavior. In contrast to speculations derived from previous studies on perceptions of wilderness (Koole & van den Berg, 2005) and behavior in the commons dilemma game (Kasser & Sheldon, 2000), we demonstrated that existential threat can result in increased pro-environmental behavior given the salience of a pro-environment norm. However, as the results of Study 2 indicate, existential threat can also lead to decreased conservation tendencies when people are confronted with a salient anti-environmental norm in the situation. This qualifies two overly pessimistic positions on the relation between terror management and pro-environmental behavior. One holds that due to the association of nature and death, attitudes towards conserving the natural environment become inevitably negative under conditions of existential threat (Koole & van den Berg, 2005). The other asserts that mortality salience always increases greed and thus impairs cooperative management of common natural resources (Kasser & Sheldon, 2000). Instead, we found that existential threat can even promote conservation when respective social norms exist and are salient. As these pro-environmental norms are increasingly spreading in light of global problems such as climate change, this finding might give rise to the optimistic notion that self threat promotes, rather than prevents, pro-environmental action. This might be indicated by the main effect of mortality salience on initial resource use intentions in Study 2.

Our results show that pro-environmental behavior can serve a terror management function. Of interest, combining this evidence with recent findings by Vess and Arndt (2008) leads to the conclusion that both anxiety buffers proposed in TMT can be involved in this effect independently. Whereas the results by Vess and Arndt show that MS can increase environmental concern when environmental action determines people's self-esteem, our present findings indicate that people exhibit pro-environmental behaviors when they consider it part of the cultural worldview. That is, pro-environmental cognitions and behavior might both bolster people's self-esteem and validate their cultural worldviews under conditions of existential threat.

However, one may ask whether the reported effects of mortality salience may generalize to other kinds of self-threats. Recent research on whether explanations of MS effects on cultural worldview defense should be restricted to self-preservation related

Table 3
Relative intention to choose reusable cups and actual choice as a function of mortality salience and norm salience in Study 3.

	Mortality salient		Mortality not salient	
	Pro-environmental norm salient	Environmental norm not salient	Pro-environmental norm salient	Environmental norm not salient
N	29	29	28	21
Intention to take reusable cup	$M = 2.44$ $SE = 1.22$	$M = -3.76$ $SE = 1.21$	$M = -1.80$ $SE = 1.34$	$M = -1.54$ $SE = 1.40$
Proportion of reusable cups taken	$M = .58$ $SE = .09$	$M = .19$ $SE = .09$	$M = .31$ $SE = .10$	$M = .24$ $SE = .10$

motivations or should be extended to more general motives related to uncertainty reduction and control might speak to this question. I. McGregor, Zanna, Holmes, and Spencer (2001, Study 3) as well as van den Bos, Poortvliet, Maas, Miedema, and van den Ham (2005, Studies 2 and 5) found parallel effects of MS and uncertainty salience on worldview defense (I. McGregor et al., 2001) and justice sensitivity (van den Bos et al., 2005; but for non-replications of parallel uncertainty salience effects see Routledge, Arndt, & Greenberg, 2004, Study 2; Landau et al., 2004, Studies 3 and 5; Landau & Greenberg, 2006, Studies 2 and 3; Martens, Greenberg, Schimel, & Landau, 2004, Study 3). In favor of a control restoration explanation of MS effects, Fritsche, Jonas, & Fankhänel (2008) found that MS effects on ingroup support and defense were eliminated when partial control over one's own death was made salient. In addition similar effects could be demonstrated by reminding people of the possibility of life partnership loss or of long-term unemployment. These results open the possibility that MS effects might be interpreted as effects of more general threats to epistemic certainty and control. According to these lines of thinking, greater compliance to pro-environmental social norms might be an expression of clinging to group norms in order to either reduce uncertainty about what is right or wrong or to strengthen one's connection to the social ingroup as a vicarious agent of control. In future research these possibilities should be investigated.

As predicted by our combination of TMT and norm focus theory, pro-environment norm cues increased conservation attitudes and behavior in the MS condition. However, interestingly, across the studies, no effect of norm focus was found when mortality was *not* salient. A possible explanation of this unexpected null-finding can be derived by considering both the TMT/norm focus approach and the nature of the norm salience manipulations involved in the present studies. Instead of manipulating norm salience explicitly, we used rather subtle treatments to induce a focus on pro- or anti-environmental norms. This was in contrast to injunctive norm treatments used in studies about littering by Cialdini, Reno, and Kallgren (1990) who put an explicit normative message under the wiper of drivers' cars concluding with the appeal, "Please, do not litter!" In other studies Cialdini et al. (1990) also used more subtle norm salience manipulations involving a confederate who crossed drivers on their way back to their car in a parking lot where s/he either littered or not. Although rather implicit, this procedure involving face-to-face interaction is likely to have elicited much more attention than reading the introductory paragraph of a paper-and-pencil study (Study 2) or listening to some minor remarks of a lay actor in a video (Study 3). Only the littering experimenter manipulation in Study 1 comes close to the parking lot confederate manipulation and interestingly, in this study, norm congruent behavior was indeed increased in the MNS condition at a descriptive level. However, only medium relatedness between primed norm (anti-littering) and measured pro-environmental attitudes (evaluation of low pollution car) and topic of information search (marine ecology) might have prevented significant norm salience effects to occur in Study 1. In sum, the norm salience cues used in the present studies might have been too subtle or not related closely enough to the relevant behavioral norms to increase norm focus in the control conditions.

But why did these subtle manipulations of norm salience work in the conditions where mortality had been made salient? According to TMT people depend on valid cultural worldviews to buffer existential threat. Thus, it seems plausible that people might actively search for information indicating the relevant social norm in a specific situation when mortality is salient. This might be especially likely when the situation is unfamiliar such when people participate in a laboratory psychological study. Hence, MS participants might have been highly vigilant for normative cues in the experimental situations, leading to

higher readiness to interpret the subtle normative cues as an expression of the relevant cultural worldview. As a consequence, normative cues could then affect behavior when mortality was salient but were ineffective in enhancing pro-environmental behavior when mortality was not salient.

5.2. Implications for the promotion of pro-environmental behavior

The present research not only provides a new explanation of pro-environmental behavior, but it might also illuminate possible paths of promoting eco-friendly conduct in everyday life. Hereby, it goes beyond previous research on the relation between environmental threats and pro-environmental action. The first important difference concerns the origin of threat. Whereas previous approaches have targeted effects of threat which emanate from a certain state of the natural environment (e.g., urban pollution), from our perspective, those threats can also affect environmental action tendencies which do not have their origin in the natural environment. Instead, self-threats might arise from both the basic human condition (i.e. human mortality) as well from threatening societal conditions unrelated to the environmental crisis, such as economic decline, anomie, or terrorist threat. The second difference between approaches is that in previous research, threats and social norms have been treated as independent predictors of pro-environmental behavior (e.g., Bamberg et al., 2007; Homburg & Stolberg, 2006). However, the present approach suggests an interaction of both factors in determining pro-environmental conduct. This means that the presence of threat should catalyze the impact that salient pro-environmental norms have on actual behavior. This might be of importance for interventions, as campaigns aimed at strengthening pro-environmental norms might be particularly effective when they target people who experience increased levels of threat. On the other hand, if strong pro-environmental norms are already present in a society or a group of people, then it might be appropriate to highlight (existential) threats associated with the environmental crisis.

Obviously, the effect of threat depends on the immediate social context in which relevant pro-environmental norms are either present or not. To provide a context in which threat facilitates, rather than prevents pro-environmental behavior, it seems to be important that pro-environmental conduct be generally valued in society and that at the same time, these norms are actualized in the everyday life of people. The latter point means that pro-environmental norms have to be communicated in public and to be applied to the various domains of behavior. That even clinical psychologists can help to establish pro-environmental norms has recently been put forward by humanistic thinkers. Specifically, it has been argued that pro-environmental behavior can be integrated into the concept of self-actualization (e.g., Kuhn, 2001; Pilisuk, 2001) and into clinical practice (Berger, 2008).

It is often discussed whether threat appeals can help to promote pro-environmental action (e.g., Baldassare & Katz, 1992; Homburg & Stolberg, 2006). Our research indicates that a combination of threat and norm appeals can be effective. Although, it is not yet clear what kinds of threat effects our findings generalize to, pronouncing the consequences of the environmental crisis, which threaten the foundations of individual existence and security, might be a way to increase the impact of norm appeals on pro-environmental behavior. Media coverage on the devastating consequences extreme hurricanes or floods can have for inhabitants of the affected territories provide examples for how environment related existential threat might be increased.

However, for most citizens of today's industrialized countries, it should be difficult to induce *existential* threat due to environmental events such as pollution or climate change. Rather than threatening

the existence of individuals in the present, these environmental changes are most often regarded as affecting the existence of future generations and of society as a whole. This does not mean that appeals to existential threat are not applicable at all, but instead of stressing *individual* threats emanating from a worsened state of the environment, possible interventions might target threats to the *collective* existence. Evidence that collective existential threat can have similar effects on the defense of cultural worldviews as individual existential threat (Kashima, Halloran, Yuki, & Kashima, 2004) indicates that communicating the risk of collective extinction due to anthropogenic environmental crisis might strengthen compliance with salient pro-environment norms.

Although our research seems to support the use of threat appeals, the conditions under which specific threat communications might be successful are complex, and transferring our threat findings to straightforward campaigns should be preceded by further research. For example, social psychological research indicates that severe threat appeals can have rather paralyzing effects on people if at the same time they do not know how to cope (Witte, Berkowitz, Cameron, & McKeon, 1998). In addition, emphasizing the role of humans in causing threatening environmental changes implies that people *do* in fact pollute and destroy, which might undermine the belief that conservation norms are indeed unambiguously held in society (see Cialdini, 2003).

However, what our findings definitely imply for campaigning is that promoting and making salient pro-environmental social norms is a worthwhile endeavor not only in times of societal stability and security, but also – and perhaps even more so – in times of societal threat. Moreover, upholding the pro-environmental norms of society might help people to buffer individual threat collectively, as these norms indicate a collective endeavor that modern societies are facing and thus offer opportunities for individuals to contribute to a meaningful cultural universe.

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